Urdu Speech Emotion Recognition using Speech Spectral Features and Deep Learning Techniques

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***Abstract*— Speech Emotion Recognition (SER) is a process for recognizing emotions hidden in speech. The main approaches used for SER include speech signal processing which utilizes acoustic speech features. Much research is being conducted to find emotions from famous and widely spoken languages like English, German, and others. However, SER for low-resource languages is still in the growing phase. In this regard, few authors have worked on SER of low resources languages like Persian, Arabic, Urdu, Punjabi, Pushto, and Sindhi. The existing work has limitations like few publicly available datasets and a lack of robustness in their SER model. This study contributes to developing a robust SER model for the Urdu language, leveraging spectral speech features' power and the latest deep learning techniques based on 1D-CNN (Convolutional Neural Network) architecture to recognize Urdu speech emotions. This study uses the first Urdu language benchmark speech dataset, "URDU", publicly available for SER research. The effectiveness and robustness of the proposed model are proved from experiments. The proposed model based on 1D-CNN architecture achieved the highest ever accuracy of 97% compared to existing work and improved baseline accuracy for the "URDU" dataset.**

Keywords—Speech Features, Urdu, Deep Learning, 1D-Convolutional Neural Network (CNN)

I. Introduction

Speech is the natural way for humans to express emotions. It is challenging for humans and machines to classify emotions found in speech as emotions have subjective nature, and there is no specific consensus to classify speech emotions.

Emotion is generally a physical or psychological reaction to a situation. Many researchers have laid the foundations for emotion theories to understand emotions in depth. Famous researcher P. Eckman [1] categorized emotions as discrete models based on physical reactions with six basic emotions, i.e., Happiness, Sadness, Surprise, Anger, Disgust, And Fear. Lang [2] categorized emotions as a continuous or dimensional model based on psychological reaction, which contains dimensions like arousal and valence, as mentioned in the circumplex model [3]. Mainly two approaches are used for Speech Emotion Recognition (SER), namely 1) SER based on physical signals and 2) SER based on physiological signals. Primarily emotion recognition from the speech is carried out using physical signals as it is easy to find emotion-relevant features from the physical signal compared to physiological signals [4].

There are many potential applications where it is necessary to classify speech emotions. SER is used in smart monitoring of patients [5, 6], emergency call centers to observe real-time speech emotions [7], capturing user's emotional experiences about products [8], AI-based virtual assistants, counselling bots, and in HRI (Human-Robot Interaction) [9].

SER is a set of techniques that involves two basic steps to perform emotion classification from a given speech signal. First and foremost, the step is to extract the essential emotion-relevant features from the speech signal. The second step is to choose a robust classifier for SER.

In literature, various speech acoustic features can be extracted from the speech signal for SER. These speech features can be categorized as 1) Prosodic Features, 2) Spectral Features, and 3) Voice Quality Features. This study uses Spectral Speech Features, particularly Chroma, MFCC (Mel-frequency Cepstral Coefficient), Tonnetz, Mel spectrogram, Spectral centroid, and Spectral contrast to carry the essential information that helps in speech emotion classification.

Several machine learning and deep learning classifiers are used in literature for emotion classification from speech. Different classifiers have advantages and disadvantages because there is no specific rule for the choice of the classifier. However, the latest literature suggests using deep learning-based classifiers for emotion classification from speech as these are more robust in performance than machine learning classifiers [10].

This study primarily focuses on a deep learning-based robust SER model that uses spectral speech features for emotion recognition task from given speech samples.

The following are the significant contributions of this study for Urdu SER:

1) Foreground important speech acoustics features that can be used to improve the performance of the Urdu SER model.
2) Propose Deep learning-based models to identify and detect emotions hidden in Urdu Speech.
3) Evaluate the effectiveness of the proposed Deep learning-based model for Urdu SER.
4) Present a comparative analysis of the proposed model with baseline and existing works.

The remainder of the paper is organized as follows. Important related work is discussed in Section II. In Section

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III proposed methodology is discussed in detail. Section IV discusses experimental results, model performance, and a comparative analysis is made against existing work. The conclusion and suggestions for future work are given in Section V.

II. Related Work

Previous and current work on SER mainly uses machine learning classifiers for emotion classification from speech. Authors in [11] used the RAVDESS dataset comprising English speech. Authors used MFCC, Mel Spectrogram, and chroma as speech features and classified speech emotions using an SVM classifier.

Several studies are in the literature in which researchers make robust SER models using the latest deep-learning classifiers. Authors have used spectrograms as speech features and implemented an SER model using the latest deep Convolutional Neural Network on the EmoDB dataset comprising German speech samples [12]. Researchers have also used mel-spectrogram images as speech features and DNN (Deep Neural Network) classifier for SER on TESS and RAVDESS datasets comprising English speech.

Most authors have worked on English and German speech. Still, few studies have focused on Urdu speech for emotion recognition as Urdu is among resource-poor languages because there are few publicly available speech datasets for the Urdu language. This study spotlighted the research on Urdu speech as Urdu is an under-researched language; despite this, Urdu has excellent significance as Urdu is among the 10 widely spoken languages worldwide [13].

Existing work for the Urdu language includes work on Urdu-Sindhi Speech Emotion Corpus and achieved accuracy for Sindhi speech at 46.81% and for Urdu speech at 57.14% [14]. Also "URDU" speech dataset is a benchmark dataset of the Urdu Language, and its baseline accuracy was reported in [15] as 83.40% using the SVM (Support Vector Machine) classifier. Our Proposed study also uses the "URDU" speech dataset as it is publicly available. Our proposed study achieved the highest-ever accuracy of 97% using the 1D-CNN classifier.

III. Methodology

A general model for the proposed methodology is given in Fig.1. According to this model, offered Urdu SER model is given 100 speech samples of each category, i.e., Angry, Sad, Neutral, and Happy speech samples of Urdu voice form "URDU" speech dataset. Different emotion-relevant spectral speech features are extracted. The SER model faces challenges like choosing appropriate classifiers and a proper set of speech features. The proposed model uses spectral speech features to address the challenges mentioned above, and 1D-CNN is used as a classifier for SER tasks. 1D vector of these speech samples features is given to the 1D-CNN model. The proposed 1D-CNN classifier with hyperparameter optimization returned the best speech emotion classification model for Urdu speech.

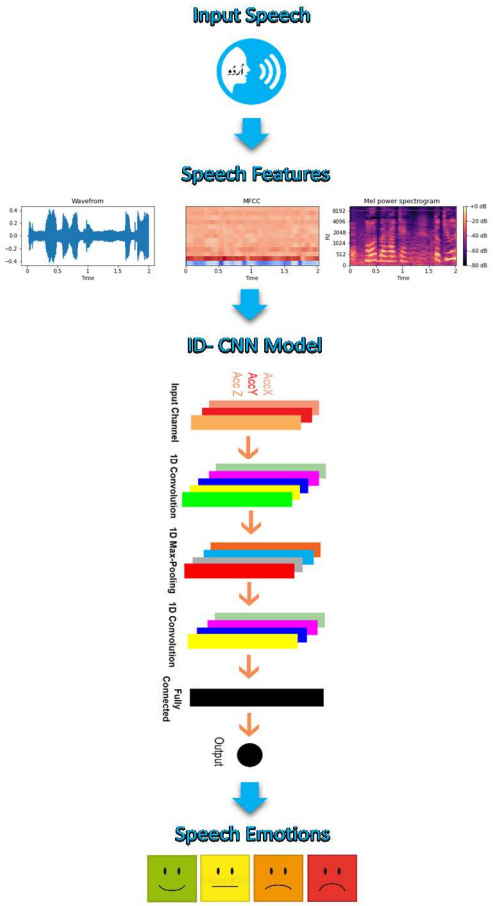

Fig. 1 Urdu SER general model

A. Dataset Description

The "URDU" speech emotion dataset [16] is the first ever publicly available Urdu speech dataset. This dataset contains natural and spontaneous speech samples from YouTube TV talk shows. In this dataset, speakers are chosen randomly. This dataset includes 400 speech samples of four basic emotions, i.e., Angry, Happy, Neutral, and Sad. The sampling rate for each speech sample is 44.1 kHz, and each is in .wav format as this is a publicly available dataset and can be used for research purposes. Following TABLE I summarizes the essential details of the "URDU" speech dataset.

TABLE I. URDU Speech Dataset Description

Dataset Name	Dataset Access	*Samples*	*Speakers*	Emotion *Classes*
URDU	Free https://github.com/siddiquelatif/URDU-Dataset	400	38 27 M 11 F	4, i.e. Angry, Happy, Neutral, Sad.

B. Feature Extraction

Feature Extraction is one of the essential steps in SER. The raw speech signal is complex to be analyzed as it may contain silence or noise at different segments[17]. The speech signal is preprocessed. Its silent parts are trimmed, and noise is removed using Wiener Filter [18]. In this study "URDU" speech dataset was ready to use as there was no preprocessing required.

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The essential spectral features are extracted from 400 speech samples in the "URDU" speech dataset. In this study given, spectral speech features are extracted.

1) Chroma is a speech feature represented as 12 element feature vector indicating each pitch class's total energy amount [19]. It also describes the intensity of each pitch class on the standard chromatic scale, and literature suggests this as an emotion-relevant feature for SER [20].
2) MFCC (Mel-frequency Cepstral Coefficient) [21] is one of the efficient speech features which can be used for SER. MFCC represents the speech signal using a filter in which phonemes (distinct sound units) are defined the same as the shape of the vocal tract (which is responsible for sound generation).
3) Tonnetz speech feature is also informative as it visualizes different speech signal notes.
4) Mel spectrogram is a form of speech signal spectrogram that shows the amplitude of speech signal mapped on a Mel scale [22].
5) Spectral centroid is a speech feature that shows at which frequency the energy of a speech signal spectrum is centered & Spectral contrast measures the difference in shape between spectra of speech signals.

Speech emotion extraction was carried out with the help of pythons Librosa library [23]. Librosa library is widely used for audio processing, speech analysis and feature extraction.

Here are the visualizations of the speech samples as waveform, MFCC and Mel power Spectrogram. As it can be observed from the speech features visualization given in Fig. 2, the Angry Urdu voice signal has patterns that are not uniformly distributed, and there are rapid fluctuations in the middle of speech segments.

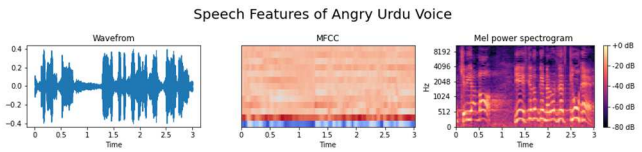

Fig. 2 Speech features visualization of Angry Urdu Voice

And in Fig. 3, the visualization of Sad Urdu voice speech features is given. Compared to the Angry Urdu voice signal, it is observed that the Sad Urdu voice signal is free from rapid fluctuations in the middle of speech segments. Sad voice signal also has long pauses or smoothness in between speech segments.

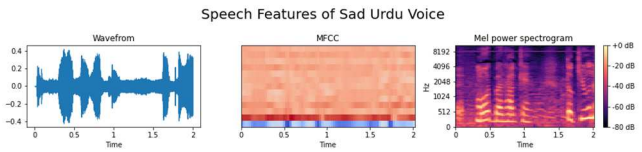

Fig. 3 Speech Features visualization of Sad Urdu Voice

Fig. 4 enunciates the speech features of the Neutral Urdu voice signal. It is seen that the Neutral voice signal is almost uniformly distributed. Unique features like Linearity and softness are also observed in Neutral Urdu voice signals.

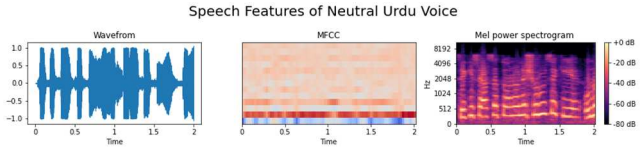

Fig. 4 Speech Features visualization of Neutral Urdu Voice

Fig. 5 demonstrates via visualization the speech features of a Happy Urdu voice; a Happy Urdu voice signal resembles with Angry Urdu voice. Both Angry and Happy speeches are full of energy. And it can also be observed that, like the Angry voice signal, the Happy voice signal also has rapid fluctuations in the middle of speech segments, and high excitement is observed in the Happy voice signal.

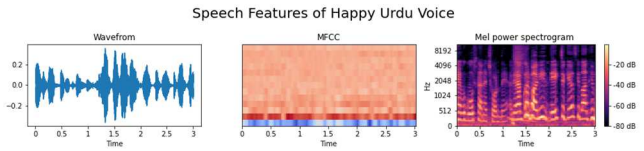

Fig. 5 Speech Features visualization of Happy Urdu Voice

C. *Model Architecture*

Initially, CNN was used for pattern recognition, image processing, and video processing, but we can also utilize a customized version of CNN for SER which is 1D-CNN model for speech [24]. From 1D it means that we are feeding our 1D-CNN model with 1-dimensional speech features vectors.

Fig. 6 shows a clear picture of the architecture of the proposed 1D-CNN-based model for Urdu SER. The proposed model is given an input speech feature set as a 1D speech feature vector. Initially, every speech feature had a different size. All the extracted spectral speech features were compressed into a one-dimensional array by taking the mean value along the time axis and then stacked on each other.

The proposed model uses this 1D array of speech features. Further convolution layers are added from which model learning parameters which helped in speech emotion classification. Here is a brief explanation of the model's internal architecture. The first two layers are Conv ID Layers of size 256, each with the same kernel size of 10. The rectifier linear units (ReLU) activation layer activates the output of both layers. After this, a max-pooling layer with a pool size of 16 is introduced. A dropout layer with a value of 0.4 is added after the max-pooling layer to prevent overfitting. After this step the model was further extended by adding two more Conv ID Layers of size 128, each with the same kernel size of 10. The rectifier linear units (ReLU) activation layer also activated both layers' output. The output of the flattening layer is received by a fully connected layer with 4 units representing 4 classes of emotions and activated with the softmax activation function. Softmax classifier is used in this classification model for predicted output. SER is a multi-class classification problem, and softmax is used for this purpose. Softmax is a generalization of logistic regression for the problem of multi-class classification. The "URDU" speech dataset was divided into a split of 70% for training and 30 for testing.

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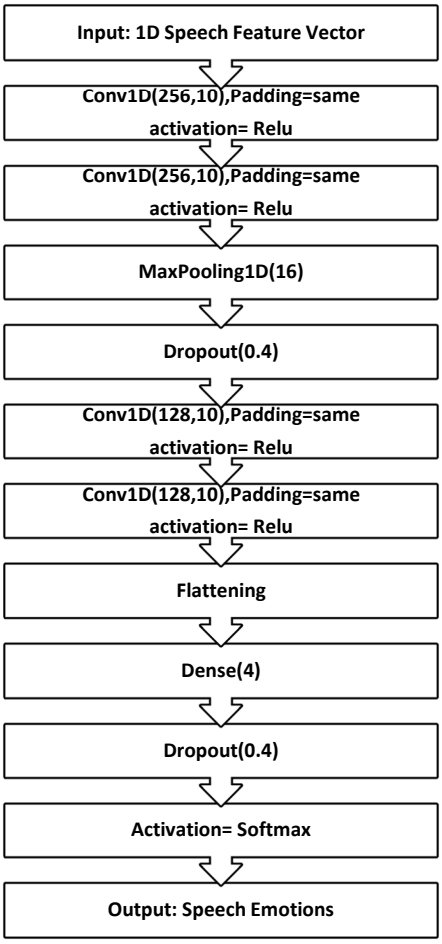

Fig. 6 Proposed 1D-CNN-based model for Urdu SER

D. *Hyperparameter Optimization*

To achieve maximum performance of the deep learning model, it is necessary to use the right combination of hyperparameter values which we generally call hyperparameter optimization. The best emotion classification model is returned by training the model with optimized hyperparameters. Given TABLE II. summarizes the hyperparameters for the proposed 1D-CNN model.

TABLE II. HYPERPARAMETERS TABLE

Type	Hyper-parameters	Value / Size
Architecture	Conv ID Layers	4 Conv 1D layers of size 256, 256, 128, 128
	Dense Layer	1 Dense layer of size 4
	Pooling Layer	1 Max-Pooling 1D layer of size 16
	Dropout Layer	2 Dropout layers, each with a value of 0.4
	Flatten Layer	-
	Kernel Size	10
Training	Activation Functions	Relu activation (after each conv 1D layer) Softmax activation (Output layer)
	Batch Size	32
	Epochs	700
Learning	Optimizer	Adam
	Learning rate	0.0001
Loss	Entropy	Sparse Categorical Cross Entropy

IV. Results & Discussion

A. *Performance Evaluation*

The performance of our proposed SER model based on 1D-CNN is evaluated based on the validation accuracy and model loss based on Sparse Categorical Cross Entropy.

Given Fig.7 visualizes the confusion matrix of model prediction about speech emotions. Where actual and predicted feelings are listed on the vertical and horizontal axis. It can be observed that the model predicted Angry and Sad speech emotions accurately as compared to Happy and Neutral speech emotions. Overall model gave an outstanding performance, and there is ignorable misclassification of speech emotions.

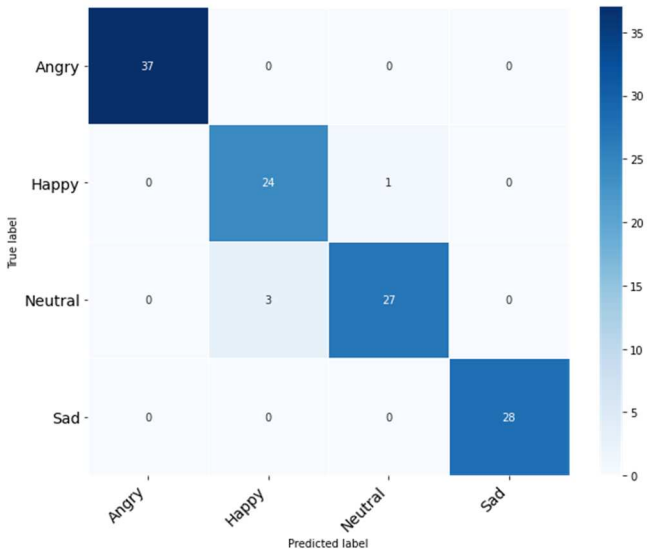

Fig. 7 Confusion Matrix for URDU dataset based on 1D-CNN model.

Given Fig.8 visualizes the model accuracy on training and test data. From this model accuracy training and test data curve visualization, it is clear that test accuracy is more remarkable than train accuracy. The overall model achieved outstanding performance. And for both training and test data, accuracy improved with no epochs.

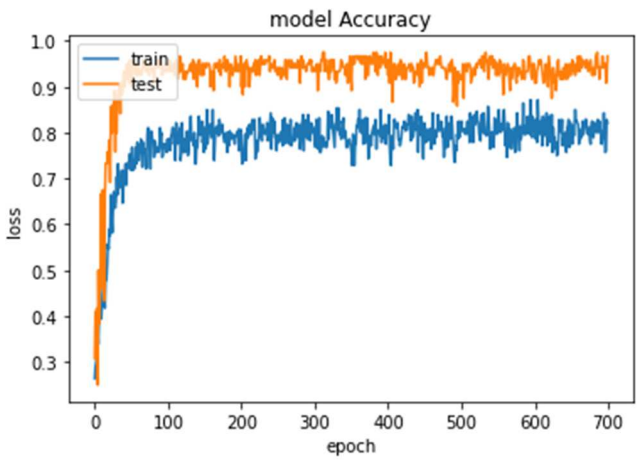

Fig.8 Model Accuracy

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Given Fig.9 visualizes the model loss on training and test data. Compared to training loss, the test data curve is close to zero when increasing the number of epochs.

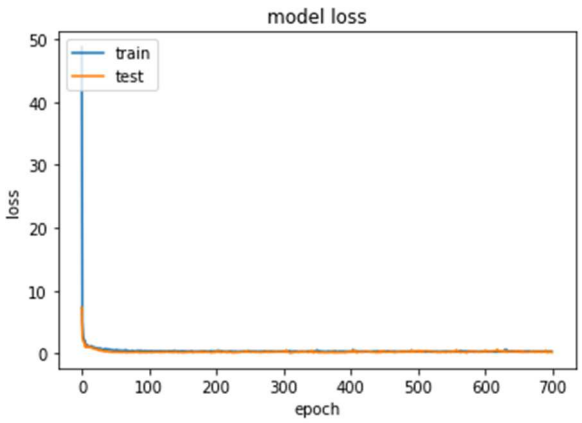

Fig. 9 Model Loss

Fig.10 visualizes the model's classification report in which precision, recall, f1 score, and support of each speech emotion category are given.

	precision	recall	f1-score	support
Angry	1.00	1.00	1.00	37
Happy	0.89	0.96	0.92	25
Neutral	0.96	0.90	0.93	30
Sad	1.00	1.00	1.00	28
accuracy			0.97	120
macro avg	0.96	0.96	0.96	120
weighted avg	0.97	0.97	0.97	120

Fig. 10 Classification report

B. *Comparative Analysis*

The comparative analysis of the proposed study is made against a baseline study and one existing work that also uses a 1D-CNN classifier.

Fig.11 visualizes the comparative analysis of the proposed study with baseline and exiting work which is made against model accuracy.

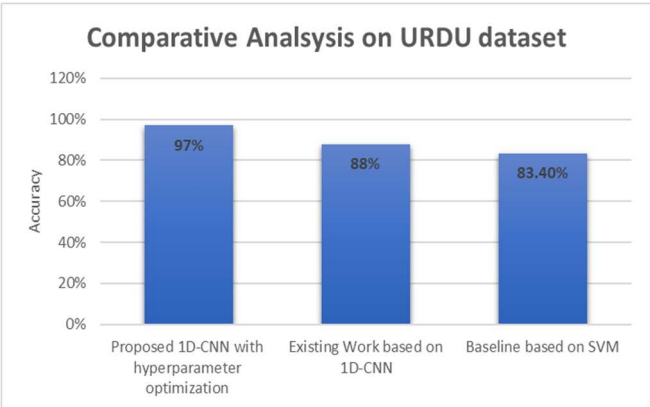

Fig.11 Comparative Analysis

V. Conclusion & Future Work

The main motivation behind this study was to create a robust SER model based on deep learning techniques to classify different emotions found in Urdu Speech. In this study, a 1D-CNN-based model is proposed to make this model robust hyperparameter optimization was performed on "error and trail". The proposed model has been evaluated on the "URDU" speech dataset. The proposed model outperformed existing works found in literature and achieved the highest ever accuracy of 97% compared to baseline accuracy of 83.40% using the SVM (Support Vector Machine) classifier [15]. And as compared to existing work given in [25], which also used 1D-CNN, achieved an accuracy of 88% on the "URDU" speech dataset.

Nevertheless, research on Urdu SER needs more focus as the Urdu language is low resource language and an understudied research area. Here are some valuable suggestions for future work, which include further investigation into given areas:

- Transfer Learning is also one of the emerging techniques which can be used in the future work. Using Transfer Learning SER data from different resource-rich languages like English, German, and Italian can be used for better results of SER, as it is proved already in some existing work that different languages have the ability to transfer learning as speech data from different languages have some common acoustic features. Many studies investigated that SER models perform well using transfer learning when given out-of-domain speech data from different languages.
- A Multimodal Approach can be further considered in the future to make SER more robust. In the Multimodal approach, not only speech data is utilized, but facial expressions and text can contribute well to revealing hidden emotion-based information that can be observed in facial expressions, speech tone, and the text uttered.
- Work on Cross-Lingual SER can be promoted to develop robust SER systems that fully understand and recognize human speech emotions. This work can be further integrated with a humanoid robot that can understand human speech emotions irrespective of language. Applications of Cross-Lingual SER include Robot Human Interaction.
- The "URDU" speech dataset is not large enough. Further, this dataset can be extended with collaborations, or data augmentation techniques can be utilized to increase the training data size, which will help improve model performance.

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